# Analytics & Big Data

Session 4: Regression I

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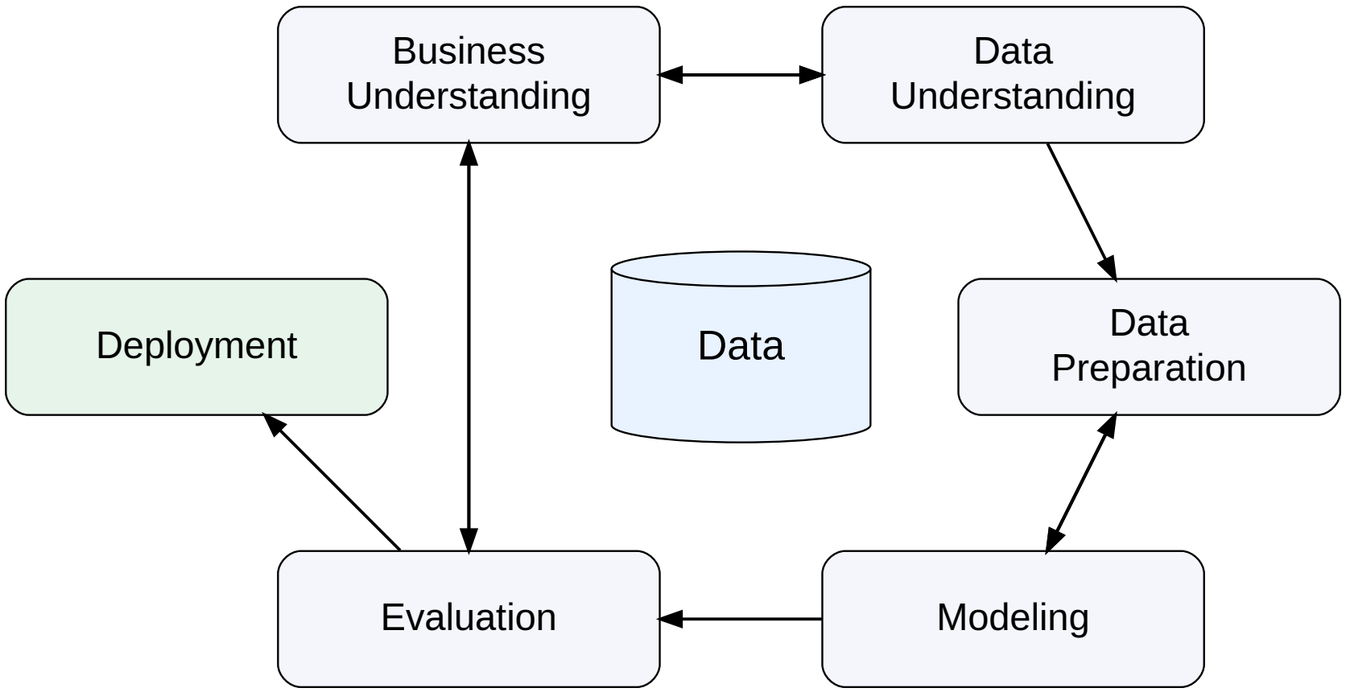
(2026-03-30)

# Learning objectives

- **Explain the stages of a model-based analytics workflow** using linear regression as an example.
- **Understand and interpret linear regression models**, including how they represent relationships between variables, how coefficients are interpreted, how OLS estimates parameters, and how model quality is assessed (e.g., via $R^2$ and overfitting considerations).
- **Describe how regression models are implemented in Python**, mapping conceptual components of the analytics process to their realization in scikit-learn.

# Process


Business Understanding
Data Understanding
Deployment
Data
Data Preparation
Evaluation
Modeling

# Business understanding

# Case: House Prices — What Drives Property Value?

Housing markets represent one of the largest asset classes in most economies. Residential real estate accounts for a substantial share of household wealth, and even small pricing errors can translate into large financial consequences.

Understanding what drives house prices is therefore highly relevant:

- **Real estate firms** use pricing models to advise clients.
- **Banks** rely on valuation models to assess collateral and manage mortgage risk.
- **Insurers and policymakers** use property data for risk assessment, taxation, and urban planning.

This case illustrates how regression analysis transforms intuition about property value into systematic, evidence-based decision support across the housing ecosystem.

# Data understanding

# Dataset: Ames Housing (Kaggle)

**Dataset:** Ames Housing Dataset

**Kaggle** is a widely-used online platform for data science and machine learning. It provides:

- Public datasets
- Notebooks for analysis
- Competitions and community discussions

**Task (5–7 minutes)**

05:30

1. Navigate to the dataset page on Kaggle.
2. Skim the dataset description.
3. Inspect the list of variables (features).
4. Open the data preview and:
    - Identify the target variable.
    - Find at least three potential explanatory variables.

Be prepared to briefly discuss:

- Which variables you expect to be strong predictors of price.

# Data preparation

…

We skip data preparation for now…

# Modeling

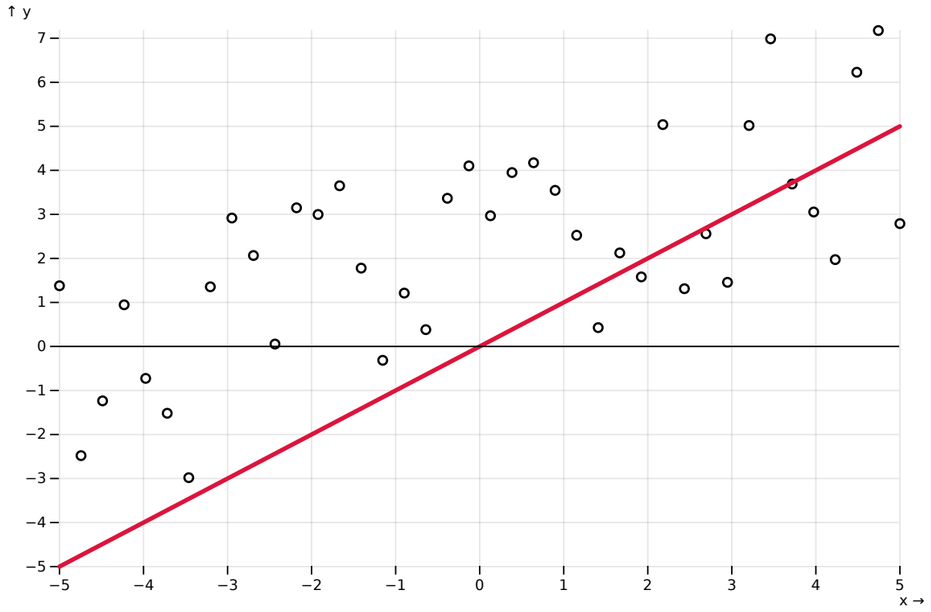
# Regression models: a visual illustration

**$\beta_0$ and $\beta_1$ move the line**

$\beta_0$ (intercept) 0

$\beta_1$ (slope) 1

Plot Data



# Model (OLS)

illustrate the optimization process

# Model results

interpret coefficients….

# Evaluation

# R^2 (does it help to predict?)

TBD: illustrate overfitting?!

TBD: other considerations? (generalization/robustness, ethics, …)

# Deployment

# TODO / TBD.

- Include a prediction example (based on a formula, python code).
- Also refer to the describe/predict/prescribe
- TBD: how can it be used to prescribe?

# Vocabulary

A **method** is a composition of formalized principles that form the basis for a stringent calculation process.

An **algorithm** is a procedure or set of steps or rules to accomplish a task. It is usually the implementation of a method. Algorithms are used to build models.

In the given context, a **model** is the description of the relationship between variables. It is used to create output data from given input data, for example to make predictions.

**Fitting a model** means that you estimate the model using the observed data. You are using your data as evidence to help approximate the real-world mathematical process that generated the data. Fitting the model often involves optimization methods and algorithms, such as maximum likelihood estimation, to help get the parameters.

**Overfitting** is the term used to mean that you used a dataset to estimate the parameters of your model, but your model isn't that good at capturing reality beyond your sampled data.

— Source: Schutt/O'Neil (2013): Doing Data Science.

# Summary

- Regression is a core example of a structured, model-based analytics process: from business question to data, modeling, evaluation, and potential deployment.
- Linear regression models quantify relationships between variables through coefficients, estimated via optimization (OLS) and evaluated using metrics such as $R^2$ while considering generalization and overfitting.
- In practice, the analytics workflow can be implemented in Python: define a model, fit it to data, generate predictions, and evaluate performance — a pattern that extends to many other modeling approaches.

Note: more on python in the exercises